



The anatomy of social capital of a Canadian indigenous community: Implications of social trust field experiments for community-based forest management

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ARTICLE INFO

Keywords:

Social capital
Social trust
Forest resources
Co-management
First nations
Economic games

ABSTRACT

The design of a community-based co-management system requires a nuanced understanding of social trust resources in the community involved. Unfortunately, however, there is a serious gap in our understandings concerning the distinct characteristics of social trust in Canadian indigenous communities. Such knowledge is essential to inform the design of management systems and processes that respect, accommodate and mobilize a community's unique culture and social resources. Our study contributes to filling the gap in our understanding of social trust patterns in Canadian First Nations, the largest indigenous population in Canada, by developing, testing and applying a systematic experimental framework for in-depth study of social trust of a First Nation community. It also explores the implications of our findings for improving the design of community-based co-management systems. We conducted our field-experiments in a community located in the northwestern forests of Ontario, Canada using a modified Investment Game (IG) as a core method in combination with survey instruments and games designed to trigger and measure behaviors related to specific motives that underpin IG allocations. Our methodology was designed to uncover fissures in the community's social trust related to gender and age. Our findings revealed that 'others-regarding' motives for trust are prevalent in the community and thus the bonding social capital is dominant and therefore social trust is rarely extended to individuals and institutions beyond the community. The findings also showed that cultural norms, especially with regard to the roles played by Elders and women as trustees and trustors in the community, do matter. We conclude with the implications of our study for the design and implementation of forest resources co-management agreements of the First Nation communities.

1. Introduction

There are more than 1.6 million people who identified themselves as Aboriginal during the 2016 Census of Canada. The First Nations of Canada are Canada's largest indigenous population and have a special role to play in forest management. The aspirations of many of the First Nations to benefit and take control of their forest resources are being encouraged by courts' decisions as well as negotiated agreements with the relevant governments of Canada. Many indigenous groups (First Nations) have entered into cooperative resource management – including Forest Co-management - agreements with many provincial governments.¹

First Nations' people are well known for their traditional wisdom based on living in harmony with nature and seeing all things as interconnected (Mussell et al., 2004). This can often lead to conflicts and disagreements between western worldviews and their own. A lack of understanding of the cultures and social resources (social capital) of First Nations people by external advisors, experts and regulators who influence the design of the co-management system and/or participate in its management, often results in a failure to mobilize the communities' social capital and causes costly conflicts (O'Flaherty et al., 2008; McGregor, 2000; Castro and Nielsen, 2001). A deeper understanding of the social resources of the First Nation communities may make an important contribution in reducing conflicts and improve the

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¹ There are some co-management agreements with the Federal Government of Canada also but these are not focused on forests and are broader such as James Bay and Northern Quebec Agreement focused on Fish and Wildlife Regime and Inuvialuit Final Agreement focused on Wildlife and Environmental Management Regime.

effectiveness of forest co-management agreements in Canada (RCAP, 1996a, 1996b; McGregor, 2011).

A core resource of social capital is social trust. Social trust is reflected in the willingness of individuals within a social unit to accept their vulnerability to unknown others (Mayer et al., 1995; Leana and Van Buren, 1999; Welch et al., 2005). Social trust facilitates collaboration and the development of strong social networks that positively contribute to the achievement of collective goals. There is a growing literature of forest co-management systems highlighting social trust (capital) as a key success factor in the design and implementation of such systems (e.g., Guillén et al., 2015; Kahsay and Bulte, 2019; Ido, 2019). Many studies (including Dahal and Adhikari, 2008; Lee et al., 2017; Roslinda et al., 2017) have examined the role of social capital in CBFM systems. Social capital was measured in a variety of ways in these studies, with the most common measure being social trust. The design of community-based co-management systems requires a nuanced understanding of social trust resources in the community involved. Unfortunately, there is a serious gap in our understanding of the distinct characteristics of social trust in First Nation communities. Such knowledge is essential for the design of management systems and processes that respect, accommodate and mobilize a community's unique culture and social resources.

One of the major gaps in these studies is the absence of explicit consideration of the different motives that underpin the development of social trust and their implications. While the conventional definition of social trust attributes the acceptance of vulnerability to unknown others as stemming from the assumption that others are likely to respond to it without exploiting it opportunistically, experimental evidence shows that the acceptance of vulnerability to unknown others (i.e. 'trusting'), may be based on a variety of motives (Lewis and Weigert, 1985; McAllister, 1995). The common sources of the social trust may differ between individuals in the community, the region or the nation. For example, individuals may decide rationally that the expected costs of monitoring or lost opportunities as a result of distrust are higher than the expected costs of exposure to opportunism that accrue from trusting unknown others, on the basis of their past experiences in the community they live in. Others may trust strangers as a consequence of living in communities where the social norms require trusting everyone, or all members of their group. Social trust may reflect unconditional generosity toward others and even a preference for taking risk or love of gambling. While types of social trust that emerge from different sources (motives) may lead to different consequences in some instances, in many instances different types can serve as substitutes to each other in providing lubrication to some social processes (Putnam, 1993), reducing transaction costs (Arrow, 1972), supporting collaboration (Gulati, 1995; McAllister, 1995) and facilitating information flows (Uzzi, 1997; Butler Jr, 1999).

The type of motives that leads to a display of social trust determines whether it contributes to bonding (strengthening bonds and networking in the community) or bridging (be open to help connect with outsiders). Thus, it also indicates whether the social trust is likely to be extended to persons and institutions beyond the community and whether it persists (e.g., is it resilient to threats, erosion over time, or incidents that contradict prior beliefs). The motives also indicate whether individuals are inward looking and conservative, or outward looking and open to innovation.

The second gap is related to the lack of studies focused on Aboriginal peoples of Canada and studies incorporating specific cultural features of these peoples. In the Canadian Aboriginal culture, elders and women are granted a special status due to various reasons (discussed in Section 2), and therefore it is possible that the motives of social trust and their implications vary across men and women, and elders and non-elders. We could not find any study that addresses these aspects either for the Canadian Aboriginal people or other indigenous cultures.

We address these two gaps. Our study has three objectives: 1) to develop a method to measure and analyze social trust patterns and motives of social trust in Canadian First Nations; 2) to assess the

composition of the variety of motives (sources) driving trusting behaviors that determine the nature of the social capital of the aboriginal community, and explore internal fissures of social trust by investigating the impacts of the provision of partial information about identities on allocations across gender and age groups; and 3) use our findings to draw implications for the design and implementation of community-based co-management systems of forest resources in First Nations communities.

To accomplish our objectives, we developed and tested a research framework that includes field-experiments and structured surveys. The framework was designed to provide fine-grained data about the level and distribution of social trust in a community by the motives that induced individuals to accept their vulnerability to unknown others in the community (i.e., to trust them).

In the design of the field experiments to generate data for our study, we used the Investment Game (IG) (Berg et al., 1995) as a core instrument. We also conducted experiments with other games (the Dictator and Risk Games) designed to trigger and measure behaviors related to specific motives that underpin IG allocations by trustors. These allowed the decomposition of IG allocations to reveal the principal motives that trigger them. Some information about motives was also obtained through surveys. Since the design and implementation of forest resources co-management also requires information about fissures in trust patterns (if any) within the community,² we included manipulations in our field experiments where partial information about trustors and trustees' identity (age group or gender) was revealed.

In executing the study, we encountered some theoretical and empirical issues that required resolution. The main theoretical issue was how motives are processed by trustors in the game. We considered two theoretical perspectives. The first (the economic rational model) assumes that the choice of trustors is based on a multi-dimensional integrated utility function where decision makers (the trustors) consider several motives simultaneously in assessing their allocations, assigning each objective a weight representing its importance. The second (a bounded rationality perspective) recognizes the complexity of the assessment and assumes that decisions are based on simpler heuristics where attention by the trustor is paid only to the dominant motive for the particular choice context. We tested both models.

The remainder of this paper is organized as follows: In the next section, we provide the context of our study – First Nations, Forest Management and women and elders in aboriginal culture. In section 3, we review the literature related to the nature of social capital and trust, and studies on Social capital and CBFM systems. We follow in section 4 with a description of the study site and the experimental and analytical methods used. Results are presented in section 5, and a discussion of their implications to CBFM systems follows in section 6. The conclusions of the study and its contributions are presented in section 7.

2. Context

2.1. First nations and forest management in Canada

About 94% of Canada's forest land is publicly owned but a large portion of it is potentially subject to aboriginal title, and it is managed by provincial, territorial and federal governments. These forestlands are commonly called Crown forests. The Canadian provinces have their rules and regulations related to forestry but in general these forest lands are managed and harvested by private companies under provincial governments; licensing systems commonly called forest tenures. These

² Heterogeneity of social groups and its implications for community-based forest management have been discussed by many scholars including Poteete and Ostrom (2004) and Adhikari and Lovett (2006). However, none of the studies have discussed implications of heterogeneity in the motives of trust to CBFM. We are testing it first time.

licenses are designed to ensure long-term timber supply to forest industry while keeping the government control on logging operations and forest management practices (Burton et al., 2003). The timber licensing systems were modified regularly to incorporate the dynamics of scientific forest management and the emergence of free market and free trade paradigms.

In brief, till the early 1990s, forest management was dominated by rational forest planning and sustained yield timber production (Burton et al., 2003; Kant, 2007; Wyatt, 2008). The conditions of forest tenures, prior to 1992, were generally negotiated between the provincial government and forest industry, and First nations peoples were excluded from these forest tenures and forest management systems (Ross and Smith, 2002; Wyatt, 2008). However, aboriginal peoples continued to practice their traditional land use activities on these forest lands while not having any opportunity to participate in forest management decisions making (Wyatt, 2008). In the light of continued pressure from Aboriginal peoples and some recognition of the existing aboriginal and treaty rights prior to colonization by the Supreme Court of Canada, the federal government signed some resource agreements with the First Nations. The James Bay and Northern Quebec Agreement (1976), which did not include forestry, attempted to replace aboriginal title by a co-management regime. In the 1990s, strong opposition by the Cree (aboriginal) people to the large-scale harvesting on their land leading to a legal action resulted in a new agreement with the provincial government of Quebec in 2002, which included forestry but failed to address the issue of aboriginal title (Wyatt, 2008). Similarly, the Nisga'a aboriginal people of British Columbia, after a long struggle with the provincial government related to forestry operations on their lands, signed an agreement in 1998 with the federal and provincial governments; the agreement gave them the control of their traditional territory but was subject to a number of conditions imposed by governments (Wyatt, 2008).

In 1992, after the Rio Summit, forest management started moving from sustained yield timber management to sustainable forest management, and the provincial governments modified their forest management policies and forest tenure systems accordingly (Kant, 2007). The current tenure systems have opened new opportunities for Aboriginal peoples' participation and involvement in forest management practices (NAFA/IOG, 2000; Wilson and Graham, 2005). The two key changes are the mandatory requirement of consultation/participation of aboriginal peoples in forest management planning process (which is also the requirement of FSC Forest Certification) and access/allocation of timber/forest tenures to First Nations. However, even in these new arrangements, aboriginal peoples are asked to accept the dominant western scientific forest management system and economic system, which are not in consonance with the aboriginal views of forest management and economic development.

It is important to note that no resource management system and economic system, including the dominant western science-based forest management system and free market based economic system, can be neutral or objective; every system is subjective and is the product of the world view, society and culture of the group that produced it (O'Flaherty et al., 2008; RCAP, 1996a; Stevenson, 2005). Hence, no resource management system and economic system can be objective and neutral to two societies or cultures that are quite different (LaDuke, 1994, p. 146), and be applicable and acceptable across diverse cultures.

Irrespective of these challenges, different types of regimes of aboriginal involvement in forest management have emerged in Canada. Wyatt (2008) grouped these regimes in five categories: (i) forestry excluding First Nations; (ii) forestry by First Nations - First Nations are part of the existing forestry regimes; (iii) forestry for First Nations - forest managers try to incorporate aboriginal values and forestry knowledge in the existing forest management systems; (iv) forestry with First Nations - aboriginal peoples are equal partners in forest management; and (v) aboriginal forestry - a new form of forestry that is based on aboriginal rights, values and institutions but uses knowledge and

techniques from both systems - western science and traditional ecological knowledge. Other scholars have also suggested the co-existence approach, which requires establishment of Aboriginal systems of forest management rather than integrating aboriginal systems into mainstream systems, for resolving ongoing forest management issues, (McGregor, 2011) which is similar to the concept of Aboriginal Forestry by Parsons and Prest (2003) and Wyatt (2008).

These are incremental steps in the right direction, but forest management systems, which will be some version of co-management due to the current reality on the ground, should embody the principles of Aboriginal/non-Aboriginal relations proposed by the RCAP. In the view of the RCAP Commissioners (1996b), the theory as well as the practice of forest/resource management is problematic. The RCAP (1996b) observed:

"While many employees of resource management agencies know that Aboriginal people living on reserves continue to harvest on Crown lands, they are generally unaware that most do so in accordance with their own rules of common property. Nor are they aware that Aboriginal people generally consider state rules an unfortunate imposition. In part, this is a reflection of the way those agencies are structured. Authority is centralized and flows from the top down, and the environment is reduced to conceptually discrete components, such as forests, parks, fish and wildlife, that have traditionally been managed independently (although less so as governments commit to principles of sustainable development or holistic management)."

These observations should be the foundation of co-management agreements.

2.2. First nations culture, women, and elders

The people of about 617 First Nations are indigenous people of Canada representing about 50 cultural groups (Aboriginal Affairs and Northern Development Canada, 2013a). These communities were independent nations at the time of forging military and trade alliances with colonizing European nations, and therefore are called Nations. At present also, in terms "their coherence, their distinctiveness, and their understanding of themselves" they are Nations (Aboriginal Affairs and Northern Development Canada, 2013b, 15). The cultures of most of these First Nations have some common key values, beliefs and practices. The common worldview of the First Nations is based on having a respectful, harmonious, and reciprocal relationships with nature, non-possession, and collectivism which is in contrast to the Western Worldview characterized by control over nature, individualism, and materialism (Mussell et al., 2004). In First Nations cultures, women and elders are treated quite distinctively, respectfully, and honorably.

In the First Nations cultures, women are the foundations of families and communities. Generally, when men are out in forests to gather, hunt and fish for their food, women manage the home, the family, and the community. Women are also responsible for teaching beliefs, values, customs, religious ceremonies, and languages to children. The new relationships of a couple are approved by the mother or the oldest woman in the household, and women teach social norms and practical aspects of a married life to a new couple. In these cultures, it is also believed that the supernatural being or the creator embodies the spirit of water. Women carry their child in water for 9 months, and at the time of delivery, this water flows from within them (Anerson, 2010). Hence, water is treated as the connection of women to the Mother Earth. It is women, not men, that are then responsible for the water at all cultural ceremonies (First Nations Voice, 2010). In view of all these special roles and contributions - their spiritual place as the Life Giver and different roles in sustaining their culture and future generations, women hold status of honor and respect (First Nations Voice, 2010).³

³ The discussion in this paragraph is based on First Nations Voice (2010).

The word “elder” has a very different meaning in these cultures. An elder does not mean an elected or nominated position like the Chief of a nation/ the Mayor of a city. It is also not a self-appointed position. There may be many elders in a First Nation. A person gains the status of an elder from the people of a First Nation when they achieve certain attributes that are recognized by the members of the nation as meriting the status of an elder. These attributes include knowledge of all the culture, language, norms, teachings, traditions, songs, and respectful relationships with the Mother Earth; strong adherence of the nation’s cultural practices in daily life; and voluntary dissemination of the knowledge, practices, and wisdom to other peoples. The elder should be able to communicate - explain knowledge and experiences - to others at their individual levels of understanding. Similarly, an elder should be able to tell a story in a way that it seems relevant to the life and experiences of a listener, but at the same time remains within the broader framework of the nation’s culture (Stiegelbauer, 1996). In the words of an Ojibway traditional teacher:

“He’s one who knows the teachings, who knows them so well that he’s able to live by them. ... and now he can give people those teachings and not only can he give them those teachings, he can help them to understand them because he’s lived it.” (Ojibway Traditional Teacher, Sudbury, Ontario [(Stiegelbauer, 1990)])

In brief, elders are respected not for their age but for their knowledge and teachings about “being a human being” and their worldly wisdom. Their teachings are seen as essential for a strong sense of cultural identity and healing (Stiegelbauer, 1996).

In brief, elders and women are treated as the keepers and teachers of aboriginal worldviews and Traditional Ecological Knowledge (TEK), and therefore play a critical role in all community-based resource management and other community-focused decision making (Stiegelbauer, 1996; McGregor, 2000).

Given these special places of elders and women in the community, we expect that elders and women are likely to enjoy high levels of trust from all groups in the community and can and are willing to play an effective role in overcoming internal fissures in social trust across gender and age-groups.

3. Social capital and community-based forest management: review of the literature

The modern usage of the term social capital emerged in the ‘60s in the context of community studies (Jacobs, 1965). It was used to describe collective resources that are key to the survival and functioning of networks of strong interpersonal relationships in urban neighborhoods, that evolve and provide the basis for collective action within them (Jacobs, 1965). In the 1980s, the concept of social capital was subject to several theoretical sociological analyses, the most prominent among them were the contributions of Bourdieu (1977, 1986) and Coleman (1988). Both Bourdieu and Coleman recognized that social capital may have both negative and positive consequences, for example, strengthening bonds between members of the community that increase collaboration and reduce transaction costs while at the same time limiting the openness of the community to flows of information from outside the community and thus constraining innovation (Nahapiet and Ghoshal, 1998). Coleman’s work emphasized more the benefits of social capital and influenced Putnam’s (1993) path-breaking study of the antecedents of social capital. This study led a resurgence of interest in the concept of social capital and its relationship to economic and social development. The study revealed positive relationships between the wellbeing of citizens and persistent differences in social norms, the intensity of citizens’ participation in voluntary organizations, and the social trust that such participation generated. The definition of social capital that emerged as a result of that study saw social capital as a social resource represented by the quality of connections among individuals that supports the development of social networks and the norms of reciprocity and

trustworthiness and trust that arise from them (Putnam, 1995).

The type of trust that is tied most closely in the literature to social capital formation is social trust (i.e., trust in strangers). Putnam’s work triggered a large number of follow-up studies that examined the relationships between social trust and, more broadly, social capital to the social and economic performance of communities, regions and nations. Social trust has been identified as a critical factor for the success of societies in general (Sullivan and Transue, 1999) and for the economic growth of countries/regions (Fetchenhauer and Van der Vegt, 2001; Knack and Keefer, 1997; Putnam, 1993).

Bourdieu and Coleman considered social capital at an individual level while Putnam treated at community level — Putnam’s original conceptualization was the relationships at a horizontal level but later expanded it to connections and interactions between heterogeneous groups at different levels (Putnam, 1998). Social capital at different levels has been categorized into bonding social capital - connections to homogenous people like family, relatives, kinship; bridging social capital - connections to demographically heterogeneous people; and linking - connections with people in power (Woolcock and Sweetser, 2002).

Scholars have explored the roles of social capital in environmental conservation and sustainability in general (examples include Atshan et al., 2020; Tam and Chan, 2018), and more specifically in community-based forest management systems (examples include Bizikova et al., 2012 and Gorriz-Mifsud et al., 2016). The dominant common arguments of these studies are that social capital promotes community’s participation in collective enterprises, facilitates decentralization, improves economic efficiency and governance, and increases social-cohesion. It influences forest management through different mechanisms including peer monitoring, reciprocity, and risk pooling. (Gorriz-Mifsud et al., 2016). Members of high social capital/trust communities care about their social image, act to meet community expectations, and believe that others will take pro-community actions (Atshan et al., 2020; Tam and Chan, 2018). Therefore, generally, community-based forest management (Ido, 2019; Guillén et al., 2015) and biodiversity conservation programs (Pretty and Smith, 2004) are successful in such communities.

In terms of community-based forest management systems and social capital literature, the consequences of social capital and its impacts on forest management have been the dominant theme. Weyerhaeuser et al. (2006) suggested that for community forestry in southwest China, investments in social capital linked to the building of more formal institutions are essential. Dahal and Adhikari (2008), based on a CBFM Case Study from Philippines, suggested that the key success factor for CBFM in indigenous communities is bonding social capital, but the mobilization of bridging and linking social capital are equally important because they mobilize external resources and promote bonding social capital. Lee et al. (2017) observed that the role of procedural knowledge of CBFM and forest farmers groups should be considered in enhancing the value of social capital for the sustainability of Indonesian CBFM. Ido (2019) reported that existing social capital such as social networks and cooperative norms between Community Forestry Group members of Cambodia, which were depleted by prolonged armed conflicts, supported the organizations of collective action, and forms of social capital were different for different types of collective actions. Roslinda et al. (2017) reported strong correlations between social capital (trust, social norms, and social networks) and community forestry in the Nusapati village in Indonesia. Guillén et al. (2015) studied social capital in small scale forestry in southern Sweden and found large differences in forest owners’ trust toward the Swedish Forest Agency (SFA) and the forest owner association (FOA) Södra. The study highlighted the importance of personal relationships and bonding social capital in order to understand local forest management situations.

Some scholars explored the relationships between social capital and forest governance. Gorriz-Mifsud et al. (2016) proposed a dynamic and nested model of the co-evolution of social capital and forest governance. The model identified cognitive dimensions of social capital as triggers of governance reforms; networks shaping information flows, power

relationships, trust among actors; and trust and social sanctions impinging on the enforcement of (in)formal norms and rules. Borg et al. (2015) conducted a social network analysis of forest biodiversity collaboration in Central Finland. They found that short-term governance networks operate on trust, and exchange and flow of information are important for network functions, though a network may be successful despite difficulties in flow of information.

In terms of social capital and forest management in general (not CBFM), Smith et al. (2012) have investigated the impacts of the strength of social trust and the sources of the trust on preferences of residents in 3 small southern US communities adjacent to public forests, among alternative forest management plans designed to achieve distinctly different outcomes. They have found that high levels of individuals' social trust influence their valuation of potential forest management outcomes. Their data suggests that affective or reciprocity-based ties carry substantially more weights than associative or exchange based ties in individuals' decision making. A recent paper by Kahsay and Bulte (2019) provides new insights about the relationships between the strength of social trust and the degree of formalization in forest management (specifically forest monitoring and extraction). The study highlights the importance of culture (in particular its reflection in generalized trust) in shaping the nature of institutions that regulate forest management. Using data of trust and the degree of formalization of forest management activities from groups of forest users in Ethiopia, the study established empirically a causal negative relationship between the levels of trust in the groups and the degree of formalization and the extent of the rules regulating monitoring their extraction activities. The study found that formal institutionalization and trust are substitutes, and that both approaches to forest management appear equally effective (Kahsay and Bulte, 2019, p119). We use some of these insights about the relationships between the degree of intensity and the nature of the sources of social trust and forest management to help draw the implications of our study of social trust on a First Nation community in section 6.

In addition to this review, Appendix I provides a review of key developments in the literature that emerged following the first publication of results of the Investment Game (IG) by Berg et al. (1995), a key instrument (though modified) in our empirical framework.

4. Methods

We describe the steps we have taken to achieve the objectives of this study, including ones we have taken to validate our methods.

4.1. Study site

The study was conducted on a First Nation living on a reserve⁴ located in northwestern Ontario, Canada. The total population of the First Nation is about 1500 and approximately 55% live on reserve. The traditional language of the First Nation belongs to the Algonquian language family. All members can read, write, and speak English. The total area of the reserve is about 3000 ha that is dominated by large stands of birch, jack pine, and cedar, an abundance of beaver, lynx, marten, mink, moose, muskrat, and rabbit, and many species of fish including perch, pike, trout, and whitefish in the lakes, rivers and streams on the reserve.

The Nation is economically progressive, and encourages, supports and promotes local economic development. The community is accessible by road and is only about 30 km from a non-reserve town. Traditional ways of living of this First Nation's people, such as berry picking, fishing, hunting, gathering of other edible food items, and participation in spiritual and social gatherings, remain common among community members. The standard of living of the First Nation studied is average

among First Nations in Canada. The Community Well-Being (CWB) index of the First Nation was 58 compared to 57 for all Canadian First Nations, but much lower compared to the average CWB of 77 for non-Indigenous communities of Canada (INAC, 2010). Only people living on the reserve participated in the study. The relevant community and participants' statistics are discussed in Section 5.

4.2. Experimental methods

The core of our empirical framework consists of 3 games: the Investment Game (IG), the Dictator Game (DG), and the Risk Game (RG). The IG and DG were modified to include experimental interventions that provided partial identity information to first players about second players in addition to experiments conducted under anonymity conditions. In addition, all participants in the study filled a survey providing social, economic, and demographic information and a set of questions related to trust and other measures at community-level. Those participants that agreed to participate in the games were asked to indicate their principal motives for each of their allocation decisions in the games after making the decision. Below we provide detailed descriptions of the designs of the experiments, their organization and the procedures of their implementation.

We used a within-subjects design because it controls for individual differences and requires a smaller pool of participants (Ashraf et al., 2006). This design also accommodated our need to incorporate individuals' motives to explain allocation in IG, and test whether motives identified by subjects indeed explain trustee and trustor behaviors in the IG i.e., the game people say they play is indeed the one they do play.

The design of our experiments, however, was subject to demand, practice, and carry-over effects. These effects were minimized by organizing the games so that each player played the roles of a trustor (first player) and a trustee (second player) at different times, not repeating the same game, and randomizing the sequence of three games. Scheduling of subjects for the various games was arranged so opportunities for exchanges of information about the games were minimal and opportunities for collusion among subjects almost nonexistent. In field experiments organized in communities, the possibility of some general information sharing is always present.⁵ In this sense, the field experiments are somewhat different than lab experiments. We have examined, however, the impact of randomness of the sequence of the games and did not find significant differences in the allocation of respective games due to randomization at a 10% significance level.

The three games were organized by the first author of this paper and a Post-Doctoral Fellow. The officer-in-charge in the band office of the First Nation was contacted for his approval. After the approval, a local assistant, recruited with the help of the band office, informed the residents about the decision of the band office, the venue, the dates, and time of the games. The assistant also informed residents that; (i) only one person from any household could participate in the games; and (ii) a participant, on arrival at the venue, would be asked to complete a questionnaire containing questions related to demographic, economic, and social aspects of his/her life and would be paid \$5 for completing the questionnaire. The individual would have an option to participate or not in the games after completing the questionnaire.

On arrival, each participant completed the questionnaire, and after that those willing participated in the games. In addition to the standard questions of gender, age, size of household, education, marital status, employment, and income, the questionnaire included questions on trust, self-trustworthiness, and other community-level social interactions. Questions on trust included the standard GSS question, a question on

⁴ These are tracts of land held in trust by the Crown for the settlement of a particular First Nation's people.

⁵ In small communities, information about arrival of external people, such as researchers from outside the community, the purpose of their visit, how long and where they will stay, what they will do in the community is commonly shared.

lending to community members, and a level of trust in the members of their own community (0 to 10 scale). All these three questions were trust at aggregate – community level.⁶ The question on self-trustworthiness was “how much do you agree with this statement about yourself: ‘I am trustworthy’- (scale of 0 to 10)”.

In the DG, each participant entered on a Proposer’s Response Sheet, his/her contingent allocation of \$20 between himself/herself and the second player for six cases: (i) Nothing is known about Player 2; (ii) Player 2 is a male; (iii) Player 2 is a female; (iv) Player 2 is a young person; (v) Player 2 is a mature person; and (vi) Player 2 is an elder. Each player also indicated the main motive for his/her allocation for each of the six cases from the following motives: (i) I want to be fair with Player 2 (*Fairness*); (ii) I want to have almost equal distribution between me and Player 2 (*Equity*); (iii) I am an altruistic person who cares for other people (*Altruism*); (iv) In general, other people care for me and so I cannot keep all the money to myself (*Reciprocity*); (v) I do not care for others (*Self-centered*); (vi) I need more money (*Need*); and (vii) Any other motive (please specify). The payment was made to the participant on the basis of randomly selected one of his/her choices made for six cases of information of Player 2.

Similar to Schechter (2007), our RG was designed to resemble the first move in the IG. In this game, each participant entered on the Response Sheet his/her division of \$20 between a risky investment and not investing. The return from investment was determined by rolling a six-sided die. The returns were: if the die lands on (i) one, the player loses the money invested; (ii) two, the player loses half of the money invested; (iii) three, the player gets exactly the same amount as invested; (iv) four, the player receives 1.5 times of the money invested; (v) five, the player receives 2 times the money invested; and (vi) six, the player receives 2.5 times the money invested. The total payment to a player from RG was equal to money not invested plus the return from money invested.

In the case of IG, the first player was informed that the second player will receive three times of the amount allocated by the first player. The first player was asked to make his/her contingent allocations for the same six cases, as in the DG, and indicate the main motive for their allocation for every case. The motives included were: (i) I trust other people and I am sure that the other player will return some good amount (*Trust*); (ii) I believe that when you take higher risk you will get higher returns (*Risk*); (iii) I do not care what the other person decides but I want to be fair to the other player (*Fairness*); (iv) I do not care what the other person decides but I want to have almost equal distribution between me and the other player (*Equity*); (v) I do not care what the other person decides but I am an altruistic person who cares for other people (*Altruism*); (vi) In general, other people care for me and so I cannot have all the money myself (*Reciprocity*); (vii) I do not care for others (*Self-Centered*); (viii) I need more money (*Need*); and (ix) any other motive (please specify).

Once all players completed the proposer’s round of the DG, on the evening of the second day, they were assigned a number ranging from 1 to 58, and each player was allocated to one of the six groups (unknown, male, female, young, mature, and elder) – allocation to gender and age groups was based on each person’s gender and age – some were

allocated to male/females groups while others to the age groups and the anonymous groups so as to have almost the same number in each of the six groups. Once each person was assigned a two-digit identification number – the first digit a serial number and the second digit a group identification letter, a second player was matched with each first player by randomly picking a chit from the bucket of chits for every player. In this random process, once a chit was drawn it was not put back in the bucket. After finalizing the matching of the first player and the second player, the total payment of each player (the portion of \$20 kept by the player as a Dictator plus the portion of \$20 allocated by the matched Dictator) was calculated, and was given to the player in a sealed envelope on the third day. In this way, each player participated in DG played both roles – dictator and second player.

The same random process, as discussed for DG, of selecting the second player was followed for the IG. However, in the IG, the second movers’ sheets were customized for each second mover by writing the dollar amount (three times the amount the first mover allocated to the second mover). The second mover was asked to indicate his/her contingent decisions about the money that he/she will give back to the first mover and the amount he/she will keep for himself/herself for all six cases – no information about the first player, the player is a male, female, young, mature, or an elder. The second mover’s sheet also included a question related to the main motive for his/her choice for each of the six cases, the motives included: (i) Player 1 sent a good amount and I just want to reciprocate (*Positive Reciprocity*); (ii) Player 1 sent a small amount and I just want to do what she/he did (*Negative Reciprocity*); (iii) Player 1 trusted me and I do not want to break his/her trust (*Trustworthiness*); (iv) I do not care what Player 1 did but I want to be fair to Player 1 (*Fairness*); (v) I do not care what Player 1 did but I want to have almost equal distribution between me and Player 1 (*Equity*); (vi) I do not care what Player 1 did but I am an altruistic person who cares for other people (*Altruism*); (vii) I do not care for others (*Self-centered*); (viii) I need more money (*Need*); and (ix) Any other motive. The payments to the first mover and second mover were made on the basis of their respective choices for the case, out of six cases, that was selected randomly during the pairing process (pairing the first player randomly to the second player).

The games were organized so that players would have no idea who was the second player, and for every first player, the second player for DG was different than the second player for IG. There was no communication among players.

4.3. Analytical methods

We used to interpret the results of our study a utility function based on the assumption that when facing complex decisions people simplify their choices by selecting motives that influence their allocation decisions for every context of their decision. In the IG the context includes what subjects know about the identity of their partner in the game, their role in the game and their own group identities and for second players the amount received in the game. The motives do vary with context and influence their level of giving in each situation. Given the central role this assumption plays in our interpretation of the IG results, we need to show that people in the First Nation indeed play the game they think (or say) they play and that motives may vary as contexts change.

We compared two models. The first model regressed first player’s allocations in IG on variables representing revealed preferences in DG (measuring OR preferences) and RG (measuring risk preferences), a trust level measure, and a number of socio-economic and demographic control variables. The second model incorporated the dominant motive of each of the first players for the allocations made in IG as dummy variables moderating the respective behavioral variables (allocation made in DG and RG and the trust level measure). In essence, the first model tested the assumptions that all subjects (first players) play an IG and the amounts sent by first players is a function of expectation of trustworthy response, risk and others-regarding motives. The second model tested

⁶ The social, economic, and demographic questions were at an individual and household level while the two questions related to trust were at community-level. In addition, there were only three questions related to trust in a set of more than 50 questions. In the games of IG and DG, the participants indicated their allocations to an individual belonging to specific group of people – male, female, young, mature, and elder peoples. Hence, we do not expect any priming effect of the survey on participants allocations in these two games.

⁷ The italicized text included in parenthesis after each motive, in the DG as well as in the IG, was not the part of original instruments of these games. This text has been added, here as well as in the copies of instruments, to refer these motives in our analysis and discussions in this paper.

the assumptions that different players may have different dominant motives when playing the IG, that they can articulate these motives and that these motives affect their behaviors (i.e., they play the game the way they say they play).

To allow comparison with prior research we framed our test in terms of the following series of hypotheses. First, we tested the following two hypotheses about the amount sent by the first player:

HF (Hypothesis for the first player) 1: Every player allocates in the IG on the basis of some combination of three motives: Expectations with regard to the trustworthiness of second player (Trust), Risk (measured by investment amount in the RG), and others-regarding motive (measured by amount allocated to the second player in the DG). To test this hypothesis, we estimated the following equation:

$$X = \alpha + \beta_1 * MTRUST + \beta_2 * MRISK + \beta_3 * MOR + \gamma_i * CV_i \quad (1)$$

HF (Hypothesis for the first player) 2: Player makes allocations in the IG following exclusively the dominant motives and the respective measures of the motive (as explained in HF1). To test this hypothesis, we estimated the following equation:

$$X = \alpha + \beta_1 * DTRUST * MTRUST + \beta_2 * DRISK * MRISK + \beta_3 * DOR * MOR + \gamma_i * CV_i \quad (2)$$

In eqs. (1,2), X is the amount sent by the first player, α is a constant, β_1 , β_2 , β_3 , and γ_i are coefficients of respective variables, CV_i are control variables (EDUCATION and SIZEHH), MTRUST, MRISK, and MOR are measures of trust, risk, and OR, respectively and DTRUST, DRISK, and DOR are dummy variables for the subject's stated prime motives of trust, risk, and OR, respectively, expressed by the first player for his/her allocation to the second player.

We tested two hypotheses about the amounts returned by second players.

HS (Hypothesis for the second player) 1: The amount returned by the second player depends on the amount received from the first player and the measure of OR motive of the second player (amount allocated by the player in DG). To test this hypothesis, we estimated the following equation:

$$Y = \alpha + \beta_1 * AMTRECD + \beta_2 * MOR \quad (3)$$

HS (Hypothesis for the second player) 2: The amount returned by the second player depends on: (i) the amount received by the second player but the fraction of the amount received that is returned to the first player will vary as per his/her motive for returning money in IG; and (ii) on the measure of OR motive of the second player (measured by his/her allocation in DG). To test this hypothesis, we estimated the following equation:

$$Y = \alpha + \beta_1 * DPOSRECI * AMTRECD + \beta_2 * DNEGRECI * AMTRECD + \beta_3 * DTRWORTH * AMTRECD + \beta_4 * DOR * AMTRECD + \beta_5 * MOR \quad (4)$$

In eqs. (3, 4), Y is the amount returned by the second player; AMTRECD is the amount received by the second player which is 3 times of the amount given by the first player (3×); DPOSRECI, DNEGRECI, DTRWORTH, and DOR are dummy variables for the motives of positive reciprocity, negative reciprocity, trustworthiness, and other-regarding motives of the second player, respectively. We did not expect any return from the second player if the amount sent by the first player is zero, but instead of setting the constant term (α) equal to zero, we tested it statistically by keeping the constant in these equations.

We tested all these hypotheses for all six cases of IG – one case of no information and five cases of partial information about the first and second players. However, before testing these hypotheses, we conducted simple statistical analyses to answer three descriptive questions: (i) Do players play the game the way they say they play? (ii) What games did they play? (iii) What are the impacts of partial information on the

identity and social distance on the allocations in IG?

5. Results

Sixty people completed the survey and 58 people, equally distributed between males and females, participated in the three games. The age distribution of participants in the games was: 24 young, 23 mature and 11 elders. The average size of the household of players was 4 people. Ten participants have a grade school education, 44 completed high school, and 4 had college-level education. Only five members had formal employment. Fifty-three of the players had an income less than \$20,000, and 5 players had income in the range of \$20,000 to \$30,000.

In the remainder of this section, we first (in subsection a) provide a description of the prime motives that first and second players in the IG indicated influenced their allocations. Next, in sub-section (b) we describe the patterns of money flows in the IG and the influence of partial information about the identity of the players on such flows. The subsection (c) describes the hypotheses testing and provides an answer to the question whether people play the game they say they are playing when playing the IG and thus testing the validity of using the elicited responses about motives to interpret our results. The key findings related to the RG and DG are included in Appendix II.

Before presenting the results, we would like to emphasize that these results are based on the data collected from the members of the First Nation subject of this study, and therefore should be interpreted in that context.

5.1. Player's motives for allocation in IG

The motives indicated by the first players for their allocation in the IG are given in Table 1. In the case of no information about the second player, 26 (45%) players indicated expectation of trustworthiness from the second player as the dominant motive while the remaining 32 (55%) players indicated other motives – 21(36%) others-regarding (OR), 7 (12%) risk, and 4(7%) self-regarding (SR) motives. In the case of first players in IG, OR motives included fairness, equity, altruism, and reciprocity while SR motives included being self-centered and having a need for money. With partial information about the second player, the percentage of players indicating trust motives decreased while the percentage of players indicating OR motives increased, and the percentage of people indicating risk motives varied in both directions.

In the case of second players with no information about the first player (Table 3), only 5% of the players indicated trustworthiness as the main motive, while 22% indicated positive reciprocity, 16% negative reciprocity, and 47% OR motives. In this case, OR motives included fairness, equity, and altruism. In the cases of partial information about the first player compared to the no information case, there were some trends but changes were not large: the proportions of those indicating trustworthiness (max value 9%) and risk (max value 14%) as prime motives for the amount of the money returned increased, while the proportions of OR motives (lowest value 40%) and negative reciprocity (lowest value 10%) decreased, and the changes in proportion of positive reciprocity were in both directions.

The proportion of first players (45%) who indicated trust as their motive, in the case of no information about the second player, is higher compared to 26% in Eckel and Wilson (2004), 23% in Johansson-Stenman et al. (2013), and 36% in Ashraf et al. (2006). However, the proportion of second players (5%) who identified unconditional trustworthiness as a prime motive to return money is between none in the case of Johansson-Stenman et al. (2013) and 9% in Eckel and Wilson (2004).

5.2. Players' allocations in IG and the impact of partial information about the identity of players on these allocations

Hoffman et al. (1996: 655) observed that “in laboratory experiments,

Table 1

First player's motives and average allocations (\$) to second players in Investment Game.

Motive for allocation		Second player unknown	Second player male	Second player female	Second player young	Second player mature	Second player elder
Trust	Number	26	16	21	15	21	20
	Average	9.81	9.69	10.48	8.67	9.95	10.65
Risk	Number	7	11	12	8	6	7
	Average	11.00	7.86*	10.42	11.13	10.00	13.57*
Others-Regarding	Number	21	25	22	25	29	27
	Average	9.48	8.88	9.27	8.04	8.83	12.07

* and ** represent allocation is significantly different at the 10% and 5% levels, respectively, of expectations of trustworthiness against other motives (t-test).

we cannot assume that subjects behave as if the world is completely defined by the experimenter". Subjects bring to the experiment their past experiences and their identities (Hoffman et al., 1996). Identities are forged in part as individuals develop their sense of self, reflecting their characteristics, experiences, social interaction and socialization processes through which they internalize norms of behavior. Such norms also influence how a person treats others given their perceived identities or specific elements of their identity (e.g., the social group they belong to, gender, age group etc.). First Nations community members have regular interactions related to economic decision making such as management of forest and other common pool resources, organization of social and cultural activities, and other community welfare projects. Age and gender form informal groupings that define important common partial identity characteristics. Hence, in our experiments, we explored the effects of identity differences or social distance between first players and second players examining how partial information about the identity of partners affects their interactions. Specifically, we compared the amounts sent and returned in the case of providing no information to cases where partial information was provided about the gender and age of the other player.

The allocations by the first player to the second player for six different cases of identity information provided are given in Table 2. In terms of the proportions of the total pie available to the first player, on an average the first player gave 48% (\$9.59) of the pie to the second player when no information was available about the second player, and it ranged from 42% (\$8.47) for young players to 60% (\$11.97) for elders; 43% (\$8.65) for males, 46% (\$9.22) for mature, and almost 50% (\$9.90) for females. The average allocations by the first players to young players were significantly lower (at a 10% level) compared to the case of no information about the second player, while the allocations to elders were significantly higher (at a 5% level).

The comparative analysis of gender shows that on an average the first player allocated significantly (at 5% level) more to females than males. All five groups of players allocated higher amounts to females compared to males, but the differences in average allocations to females were significant (at 5% level) only for allocations by males and mature

players. In the case of age groups, all five groups of first players allocated a higher amount to elders as compared to young players, and these differences were significant at a 5% level for all groups of first players except mature players.

The average percentages of amounts returned by the second player to the first player for six cases of partial information are given in Table 3. In the case of no information about the first player, the average proportion returned is about 29% which means the average return is less than the amount given by the first player. For the total sample, this is true for all groups of first players (male, female, young, and mature) except elders. In the case of elders being the first players, in aggregate slightly more than the amount given by the first players was returned. Specifically, male, young, and elder players returned to elders more than the amount given by the first players. Elders also returned more than the amount given by the first players to females, young, and mature players. Similarly, young players returned more than the amount given by the first players to females and young players.

5.3. Hypotheses testing

The details of the variables used in estimating the four models are given in Table 4. As stated earlier, we collected data through survey on three measures of trust. In the case of the first measure, the standard GSS question on trust in the community members (using Yes/No responses) was used. The second measure used a finer scale that measures the degree of trust in own community members on a scale ranging from 0 (no trust) to 10 (full trust). Our third measure of trust was based on revealed behaviors asking about the frequency (never, occasional, and frequent – scored 1, 2, and 3, respectively) of lending personal possessions.

The models for HF1 and HF2 were estimated for all three measures of trust - including one measure at a time. The coefficients of trust measured by GSS question were neither significant in HF1 nor in HF2 models. The estimated models that included trust on a scale of 0 to 10 and trust as a frequency of lending personal possessions were similar in terms of the signs and significance levels of coefficients of trust variables. However, the interpretation of the coefficient of the trust variable

Table 2

Average allocation (in \$) by first player to second player in Investment Game.

Category	Counts	Second Player is Unknown	Second Player is Male	Second Player is Female	Second Player is Young	Second Player is Mature	Second Player is Elder
Male	29	9.66	8.48	10.10 ^{##}	8.97	8.97	11.59 ^{\$}
Female	29	9.52	8.81	9.68	7.97 ^{**}	9.48 ^{\$}	12.35 ^{**\$\$}
Young	24	9.17	8.08	9.96	8.08	9.00	13.79 ^{**\$\$}
Mature	23	10.52	8.91 ^{**}	10.04 ^{##}	8.65 ^{**}	9.09 ^{**}	9.70
Elder	11	8.55	9.32	9.40	8.91	10.00	12.73 ^{**\$\$}
Total	58	9.59	8.65	9.90 ^{##}	8.47 [*]	9.22	11.97 ^{**\$\$}

Notes

* and ** represent allocation is significantly different at the 10% and 5% levels, respectively, from the allocation to the second player when the second player is unknown (paired t-test).

and ## represent allocation is significantly different at the 10% and 5% levels, using a paired t-test, respectively, when the allocations to females are compared with allocations to males.

\$ and \$\$ represent allocation is significantly different at the 10% and 5% levels, using a paired t-test, respectively, when the allocations to mature and elder are compared with allocations to young.

Table 3

Average proportion returned by second player to the first player in Investment Game (in %).

Category	Counts	First Player is Unknown	First Player is Male	First Player is Female	First Player is Young	First Player is Mature	First Player is Elder
Male	29	33.86	29.70	33.61	32.67	30.63	37.18
Female	29	24.85	24.22	27.07**	30.41	26.24	30.19
Young	24	33.90	29.95	35.28	37.46	30.94*	38.91**
Mature	23	29.48	28.47	29.56	27.74	29.17	30.96
Elder	11	32.83	30.93	34.97	40.66*	35.10	41.79*
Total	58	29.28	26.91	30.28	31.52	28.40	33.63**

Notes: * and ** represent proportion is significantly different at the 10% and 5% levels, respectively, from the proportion returned to the first player when the first player is unknown.

Table 4

Description of variables used in regression models.

Acronym of the variable	Description of the variable	Mean/SD/Range of values Only for continuous variables No Information Case*
Continuous Variables		
EDUCATION	Player's education (1 grade school, 2 high school, and 3 college level education)	1.86/0.58/0–4
SIZEHH	Size of household of player (person/household)	3.98/2.67/1–10
MOR	Measure of player's Other-Regarding motive (Money (\$) given to the second player in Dictator Game)	6.62/3.89/0–20
MRISK	Measure of player's risk motive (Money (\$) allocated for investment in Risk Game)	13.83/4.83/0–20
MTRUST	Measure of player's trust in community members (on a scale of 0 to 10)	5.45/3.31/0–10
Dummy Variables		
DOR	Dummy for OR motives of first player in Investment game (IG)	0.36
DRISK	Dummy for risk motives of first player in IG	0.12
DTRUST	Dummy for trust motives of first player in IG	0.45
DPOSRECI	Dummy for positive reciprocity motive of second player in IG	0.22
DNEGRECI	Dummy for negative reciprocity motive of second player in IG	0.16
DTRWORTH	Dummy for trustworthiness motive of second player in IG	0.05
DOR2	Dummy for other-regarding motive of second player in IG	0.47

Note: * The mean values of all variables, except EDUCATION and SIZEHH, are different for six cases, one with no information about the other player and five cases of partial information about the other player. In this table, we have provided mean values for the case of no information about the other player only.

is easier for the trust measured on a scale of 0 to 10, and therefore we present and discuss results for the models (eqs. 1 and 2) estimated using the scale trust variable.

The estimated results of the HF1 model are given in Table 5. The interesting results are that the coefficient of the measure of OR motives (allocation to the second player in the DG game) is positive and significantly different from zero in all six cases of our analysis while the coefficients of the measure of trustworthiness expectations and the measure of risk (investment in RG) are not significantly different from zero for all cases. Generally, these results may lead researchers to infer that the allocations in the IG do not depend on the stated trust and risk. However, such an inference is erroneous. These results only indicate that for the whole sample of 58 players, stated trust and measured risk do not have a significant influence on the amount allocated by the first player to the second player. Given that more than 50% players do not indicate expectations of trustworthiness as their prime motive and more than

Table 5

Results of regression models of first player's allocation in Investment Game (Number of observations for both models and for each case = 58).

Variable/ regression statistics	Player 2 Unknown	Player 2 Male	Player 2 Female	Player 2 Young	Player 2 Mature	Player 2 Elder
HF1 Model						
Constant	−0.57	3.33	3.72*	0.24	4.40**	6.77**
MTRUST	0.10	0.15	−0.05	0.13	0.02	0.23
MOR	0.49**	0.39**	0.16	0.28*	0.49**	0.41**
MRISK	0.09	−0.10	0.18	0.17	−0.08	−0.01
EDUCATION	1.90**	1.70*	0.72	1.04	1.31*	−1.18
SIZEHH	0.40**	−0.04	0.65	0.42**	−0.13	0.37
R ²	0.30	0.21	0.18	0.23	0.43	0.40
Significance of F value	0.0018	0.0318	0.0245	0.0302	0.0010	0.0001
HF2 Model						
Constant	4.31**	5.32**	4.50**	3.29*	3.73**	10.53**
MTRUST	0.31**	0.31**	0.28**	0.32**	0.44**	0.24
(x) DTRUST						
MOR	0.34**	0.25*	0.17	0.26**	0.40**	0.19*
(x) DOR						
MRISK	0.16*	0.07	0.14	0.31**	0.19**	0.22*
(x) DRISK						
EDUCATION	1.64**	1.00	1.31	1.19**	1.58**	−1.35
SIZEHH	0.09	−0.02	0.35*	0.26	−0.08	0.51**
R ²	0.15	0.10	0.14	0.24	0.30	0.23
Significance of F value	0.0947	0.1276	0.1105	0.0031	0.0102	0.0270

Note: * and ** represent significance of coefficients at 10% and 5% significance levels, respectively, based on Robust standard errors.

80% of players do not indicate risk as their motive, these results are not surprising, rather they are expected.

The HF1 model does not capture the influence of motives on allocations of the first player, and therefore we do not discuss specific results. In the HF2 model, we used interactions of dummy variables indicating whether a player selected a particular motive for their allocation in the IG (that is the game they say they play) and the corresponding behavioral/attitudinal measures derived from their behavior in independent experiments (DG, RG) or answers they provided from questionnaires they filled out (trust). Hence, this model captures the influence of a specific motive only for those players who have indicated that motive as a prime influence on their play of the IG. The results of the HF2 Model are also given in Table 5.

The significance levels of F values of the HF2 models for the cases of no information and three cases of age are below 10% while for the two models where identity information concerning gender of players was provided were insignificant. In the case of the four significant models, the model relating to the case where 'no identity information was provided' and the models relating to the cases where membership in a particular one of the three age groups is revealed, the coefficients of all three variables representing the motives – the assessment of

trustworthiness of others (Trust), Risk, and OR – are significantly different from zero, except the Trust variable in the case of the elders' model. This may be due to the very high value of the constant in the case of elders which means that on an average, players allocate high proportion (more than half) of their endowment to elders without the consideration of trust levels. In the case of the mature players as recipients' model, the coefficients of Trust and OR motives (0.44 and 0.40) are significantly higher than the coefficients of these variables in the no information model as well as the young people and elders' models. These higher values indicate that the players allocate higher proportions of their endowments to mature people, as compared to other age groups or no information case if their allocations are due to Trust or the OR motive. Similarly, if the motive is Risk, people allocate higher proportion to young people.

Similar to Ashraf et al. (2006), the values of the constants are positive and significant for all six cases, and it is more than 50% of the pie for elders. This means that the first players allocate a good amount of money to the second player irrespective of the identified motives and individual characteristics, and they derive satisfaction from this allocation due to some other motives that are not captured in our model, but may be due to their action of sharing money that they received from the game's organizers. A very high value of constant for elders confirms their special social position in this First Nations community. The first player allocation increases with the level of education in the cases of the second player being unknown, female, young, and mature. The first player allocation increases to females and elders with the size of the household from which the first player comes, indicating perhaps the special perspective about the treatment of women and elders that is acquired when living as part of a large family. Any other socio-economic variables measured through the survey, such as income and belongingness, were not significant in the HF1 and HF2 models probably due to the homogeneity of the community and thus lack of sufficient variance in these variables.

The results of the estimation of the HS1 Model are given in Table 6. In HS1 Model A, the constant, as expected, is not significant in any of the six cases; it confirms that the second players cannot return any money if the first players have not allocated anything to them and the second players are not paid any participation fee as in our IG design. Hence, we re-estimated eq. (3) without a constant term. The results are given as HS1 Model B in Table 6. The coefficient of AMTRECD ($3\times$) is positive and significant in all six cases, and the coefficient of the measure of OR motives (MOR) is positive and significant in all cases except the case where there is no information about the second person. R^2 for all cases are greater than 0.70 which means these two variables explain a high proportion of variance in the amounts returned by the second player. These results also confirm that the amounts returned depend on the amounts received as well as on OR motives of players. In fact, the coefficient of MOR is higher than the coefficient of AMTRECD (amount received) for all five cases in which partial information is available about the first player; in the case of no information, the MOR coefficient is not different from zero. These results indicate that in the case of no information about the first player, the amount returned solely depends on the amount received but as soon as there is partial information about the first player, the OR motives dominate over the amount received.

The results of estimation of the HS2 Model are also given in Table 6. In this model, AMTRECD is moderated by the four motives of the second player for returning the money. Three out of four interaction coefficients are positive and significant in all six cases, while the coefficient of the interaction of AMTRECD and the motive of negative reciprocity is positive and significant for all cases except the cases of a female or young being the first players. The coefficient of MOR is positive and significant for all cases except the case of no information provided about the first player. The results of this model help in distinguishing the proportions of amounts received that are returned by people driven by different motives for return. As expected, the coefficients of positive reciprocity are quite high compared to the coefficients of negative reciprocity in all six

cases, and the coefficients of negative reciprocity are also quite low compared to the coefficients of trust and OR motives. Generally, the coefficients of positive reciprocity are higher than coefficients of other motives except in the cases of a female or an elder being the first player. In the case of elders, the coefficient of trust is highest, and in the case of women, the coefficient of OR is the highest. The inclusion of interactions with the dummy variables related to motives identified by the players also improved the R^2 in all six cases.

6. Discussion and forest management implications

The objectives of our study were to develop, test and apply a systematic experimental framework for an in-depth study of social trust of a First Nations' community, and explore the implications of our findings for improving the design of First Nations' community-based co-management systems. We have developed a framework for measuring social trust using a sequence of experiments and questionnaires. We contended that the behavioral manifestations of social trust by individuals i.e., the acceptance of vulnerability to unknown others in the social unit (e.g., organizations and communities), may be based on different motives. We argued that the composition of individual motives that constitute the unit's social trust matters. To test our methodology, we needed to articulate the mechanisms that link the motives that shape intentions to accept vulnerability to the actions they trigger. We tested two alternative models about the nature of the decisions to accept vulnerability: 1) a bounded rationality model, and 2) a traditional classical economic model of multi-dimensional utility maximization. The bounded rationality theory argues that people simplify complex decisions problems by employing heuristics. We have assumed that the most intuitive way for people to simplify their choice processes is to focus, in each decision situation, upon a key dominant single motive, and not consider combinations of multiple motives as guides to their choices. To test this hypothesis, we have contrasted our decision model using the key motives elicited from players with a classic economic model based on revealed choices —i.e., explaining behavior on the basis of multiple motives where their weights are estimated from revealed behaviors. This test was also designed to answer the questions raised by Camerer (2003) about behaviors displayed in economic games and motivation: "What game do people think they are playing?" and more importantly "do they play the game they say they are playing?" These questions examine the external validity of the data our experiments produced. The results of our hypotheses' testing confirmed that the dominant single motive model based on players' inputs provided significantly more accurate predictions than the revealed choice multiple motives model. This confirmed also that at least in the IG, Canadian First Nation players were probably aware of their motives during the game and that their choices in the game were consistent with the motives they said guided their play (i.e., 'the game they said they played was the one they played'). We also found that many of the first players had different motives guiding them during the various experimental interventions as a result of the information provided about the group identity of second players. We have also tested hypotheses derived on the basis of prior anthropological research on First Nations cultures regarding the special status they accord to women and elders. It was clear that these and other choices in the game were influenced by traditional values rooted in First Nations cultures. While these experiments were conducted primarily to validate our approach, they also provided very valuable information about the effects of culture on decision processes in a First Nation community.

The second objective of our study was to use the methodology to study in-depth the composition of social trust in a First Nation community and to draw the implications of our results. Below we discuss our findings and investigate their relevance to the design of a community-based co-management of forest resources in a First Nation community in Canada.

There is a near-universal consensus in the literature that social trust plays a critical role in community-based natural resource management,

Table 6

Results of regression models for amounts returned by second players in Investment Game (Number of observations for all models and for each case = 58).

Variable/regression statistics	Player 1 Unknown	Player1 Male	Player 1 Female	Player 1 Young	Player 1 Mature	Player 1 Elder
HS1 Model A						
Constant	0.80	0.59	−3.29	−1.24	−0.06	−1.59
AMTREC	0.27**	0.20*	0.23*	0.21*	0.24**	0.24**
MOR	0.06	0.28	0.69**	0.63**	0.27*	0.44**
R ²	0.21	0.18	0.30	0.31	0.24	0.23
Significance of Chi Square	0.0386	0.0335	0.0062	0.0000	0.0265	0.0060
HS1 Model B						
AMTREC	0.29**	0.21**	0.18**	0.18**	0.24**	0.21**
MOR	0.09	0.31	0.50**	0.59**	0.27*	0.39**
R ²	0.71	0.74	0.76	0.76	0.77	0.74
Significance of Chi Square	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
HS2 Model						
AMTREC (x)	0.42**	0.34**	0.34**	0.27**	0.33**	0.34**
DPOSRECI						
AMTREC (x)	0.14**	0.09**	0.06	0.06	0.13**	0.09*
DNEGRECI						
AMTREC (x)	0.39**	0.32**	0.17**	0.24**	0.19**	0.47**
DTRWORTH						
AMTREC (x)	0.35**	0.28**	0.38**	0.25**	0.27**	0.29**
DOR2						
MOR	0.04	0.26**	0.38**	0.53**	0.32**	0.21*
R ²	0.84	0.85	0.87	0.82	0.81	0.84
Significance of Chi Square	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Other variables – self trustworthiness, dummy of belongingness, and gender, age (elder and mature), size of household, and education are not significant.

* and ** - significance of coefficients at 10% and 5% significance levels, respectively, based on Robust standard errors.

including co-management of forest resources, biodiversity and environmental conservation. Our study indicated that the over-all levels of social trust generated in the community were very high in absolute terms and in comparison, to levels of social trust for non-aboriginal communities reported in prior studies (Johnson and Mislin, 2011). The acceptance by community members of vulnerabilities to unknown others within their community, permits the smooth and efficient functioning of collective resource management and conservation processes run by community members. High levels of trust, apart from being an important precondition to a successful functioning community-based forest management system, have also some other implications related more directly to the success of a community-based forest co-management system. Kahsay and Bulte (2019), in their study of the relationship between trust and regulation in participatory forest management in Ethiopia, present evidence of a negative correlation between various measures of trust and the degree to which the forest monitoring of extraction is formalized. Additional analysis by the authors revealed that the relationship is causal. Groups with low trust levels implemented more extensive formal regulatory and monitoring systems. This relationship suggests that trust and rules are substitutes in natural resources management. Indeed, one of the major problems in community-based forest co-management systems in Canada is the tendency of government partners to insist on extensive formal rules and tight monitoring systems that clash in high-trust communities (such as the First Nation subject of our study) with their preferred informal monitoring systems guided by culture and traditions. The second implication of high-trust in the community is the impact it has on community preferences. High-trust communities place higher weights on long-term outcomes such as forest conservation and environmental protection (Pretty, 2003; Pretty and Smith, 2004). The major concerns of government partners often focus on the sustainability of the forest resources and the protection of the environment. Our findings suggest that there is an alignment between the objectives of the government and those of a high trust First Nation community. As long as transparency and inclusivity of the co-management system is maintained, allowing greater autonomy to the community to manage its own natural resources is likely to achieve both

the objectives of government regulators and the aspirations of the community involved.

We also wanted to identify the relative shares of social trust derived from different sources (motives) in a North-American indigenous community, specifically a First Nation of Canada. While trust generated on the basis of one motive is often substitutable for trust generated by other motives, especially in terms of ensuring efficient and smooth organizational processes, different types of motives have important behavioral impacts which may affect the effectiveness of particular system designs and management styles. Social trust based on different motives may have different direct impacts on the scope of social capital they generate (i.e., to whom the trust is extended and in what contexts) and its durability (i.e., the circumstances that lead to erosion of trust), and indirect impact on other behaviors such as openness to innovation or increases in conservatism.

Our elicitation of motives showed that given complete anonymity about 45% of participants based their decisions on an assessment of the degree of trustworthiness. The majority of the people, however, were willing to accept vulnerability to anonymous others because of OR motives, such as unconditional generosity. Once partial information about second players' identities was revealed to the first players, the proportion of first players making their allocation decisions on the basis of OR motives increased irrespective of the specific identity information they were provided about the second players. As well, the proportion of trust based on assessment of the trustworthiness of others declined. Similar to Smith et al. (2012) we argue that social trust based on affective motives such as a wish to display unconditional generosity to others in the community, or reciprocity, are associated with higher levels of trust and stronger ties in the community. These tend to be correlated with bonding preferences and activities. In contrast, motives that are based on the belief in the trustworthiness of others, or on rational premises such as the mental or experience-based comparison of the cost and benefits of trusting versus not trusting, are associative and exchange based and are correlated with lower degrees of social trust and weaker ties in the community. However, they tend to be associated with bridging capabilities and tendencies. The strong presence of OR motives

in the First Nation community studied suggests that the high levels of trust are rarely extended to individuals outside the community or external institutional stakeholders and their representatives. They also may underscore the potential of resistance to change and rejection of external interventions. “Bonding capital can create dense social networks defined by high localized levels of trust and social networks that tend to share similar values, attitudes, preferences and behavioral patterns...as a result, bonding social capital can impose strict social norms that discourage change” (Smith et al., 2012:208–209). Having a significant minority of ‘bridging’ social trustors in the community studied can provide (if they are coopted by the system) a link to external sources of information and new ideas. The problem of such cooptation lies in rising tensions and conflicts between conservatives and change seekers. It can be solved by having highly trusted members within the co-management team that can channel differences in points of view in the community into forces that ensure a productive change. Our study results have identified the groups of elders as the ones that can serve in such leadership capacities. Their legitimacy is sanctioned by traditions and, as our study results indicate, by an overwhelming trust they enjoy from all groups in the community. We also found that their behavior as trustors favored all groups in the community almost equally and generously.

As we have discussed, complete anonymity is rare in relatively small communities. Many management and design decisions concerning forest and environmental resources affect future generations and the decisions involve long horizons where there is little known about specific individuals. The knowledge available to planners about future generations consists mainly of demographic predictions (age group and gender composition). Age groups and gender constitute important identity markers. The influence of identity, a person’s sense of self, is manifested in socially prescribed behaviors and values (Akerlof, 1997). Some of these prescriptions may be universal (e.g., showing a measure of generosity to all community members) while others prescribe specific treatments of others in the community depending on their group identities (e.g., showing preferential treatment of specific groups). Group identities may change over time but in aboriginal communities they are embedded deeply in the aboriginal culture and thus are slow to change. Our experiments included interventions designed to uncover the impact on trust of partial identity information (age or gender) provided to trustors about otherwise unknown trustees. They also uncovered age and gender related fissures in the community’s social trust. Such information is important in organizing remedial bonding efforts.

Our findings related to the sources of trust and those related to the allocations to elders and women are highly significant from the perspective of forest management by First Nations. The higher allocations to elders by all groups may be due to a “matter of respect”. In many First Nations cultures, elders are respected for what they have learned, asked to teach others, and their teachings are seen as essential for a strong sense of cultural identity and healing (Stiegelbauer, 1996). Hence, the Indigenous players may be allocating higher proportions to elders out of a norm of respect⁸ to elders and by doing so they maintain that elders have a distinguished and worthy place in their own life as well as in the First Nations. Therefore, in forest management decision making and implementation of co-management agreements, the elders should be assigned appropriate roles as per their status among their own people. Arguably, co-management agreements signed by some forward-looking and economic-growth focused elected band leaders with governments may succeed less well without a strong support of the elders of the respective First Nations.

Similarly, the IG allocation results clearly captures the special role of women and the distinction between the role of elders and women. Hence, in forest management and other co-management agreements, the

special contribution of women is in part building and strengthening social links in the community and helping resolve trust fissures. These roles are essential and must be recognized, celebrated and utilized.

7. Conclusions

Aboriginal forestry – a new form of co-management systems based on aboriginal rights, values and institutions, and the incorporation of western science and traditional ecological knowledge – is expected to address fundamental issues related to aboriginal peoples and forest management in Canada. In aboriginal cultures, elders and women have a very special position. In this paper, our focus was to study the differences in trust and motives of trust across gender (men and women) and age groups (elder, mature people, and young people) and our results indicate that our methods were very successful to capture these differences. Our findings are similar to many qualitative studies which have clearly indicated that women and elders have a special place in aboriginal cultures. Our study has confirmed those differences in quantitative terms and have provided the quantitative measures of the differences in trust due to different motives expressed by participants for females and elders. This study, therefore, is complementary to qualitative studies.

The findings of our study have some important implications for community-based forest management decisions by Aboriginal communities. The findings demonstrate that social trust, a key component of community social capital, is not a homogeneous collective resource. It represents a diverse portfolio of individual motives that contribute to the development of the community’s social capital and social trust. There is substantial evidence in the literature that such trust supports the development of community ties, enhances communication, reduces transaction costs and increases collaboration, all outcomes that contribute to effective and sustainable community-based resource management. Differences in sources of social trust, however, may influence the strength, social and geographical scope and durability of the impacts of social trust on the community, and the contexts in which it is manifested and mobilized.

In the First Nation studied, a major source of social trust was motivated by OR preferences originating from social norms deeply embedded in the traditional local culture. Such trust is strong and resilient but is not extended to outsiders. Indeed, the strong linkages it fosters within the community and the interest in the community to defend its culture, impose barriers on communications with outsiders. Efforts in the past to reduce the influence of traditional cultures (often under the guise of modernization and progress), however, did not weaken social trust within the community. Our results suggest that despite attempts to impose “modern governance norms” to replace traditional ones, the latter seem to be imprinted on current practices and values. The attempts to override traditional norms seem to have increased distrust of outsiders and the institutions they represent. Our findings about the high level of social trust within the community suggest a likely preference in the community for trust based informal governance structure as opposed to one based on formal detailed regulations and monitoring that is often favored by government partners. Not surprisingly many First Nations are demanding now an increased autonomy as part of co-management agreements. Generally, the implications of our study indicate the importance of recognizing and understanding cultural differences and showing respect for them by both accommodating traditional local knowledge and utilizing the advantages that it creates. To overcome the cultural gap, increased engagement of the Aboriginal community members in the design and implementation of forest co-management agreements between Aboriginal and non-Aboriginal peoples are needed. Our study also demonstrated fissures in social trust in the community with the group of elders universally trusted, and thus suggesting that they can play (and do play) an informal mediation role. The women in the community also can play an import role in reducing intergenerational trust fissures in the community, as they enjoy higher levels of trust than men and already play a

⁸ Generally, people strive to avoid violating another person’s “definition of a situation” (Goffman, 1958; Miller, 2006) and respect the positive attributes, such as a moral and knowledge, of other person (Brown and Levinson, 1987).

significant role in promoting bonding within the community.

In conclusion, we would like to caution that our study was based on data from one First Nation and as such our recommendations may not apply to other nations. Our results are based on the members of the First Nation and should be interpreted keeping this in perspective. We are aware of the rich diversity of cultures and traditions that exist among First Nations of Canada and therefore call for studies in other First Nations. We believe, however, that some of our findings with respect to social trust levels and their sources and distribution are likely to be found in many other First nations communities.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We express our deep gratitude to the elders, leaders, and members of the First Nation who supported and participated in the study. Financial support from Social Sciences and Humanities Research Council, Canada, Natural Science and Engineering Research Council, Canada, and Sustainable Forest Management Network, Canada are highly appreciated. None of the above is responsible for any errors in the paper or opinions expressed.

Appendix I Prior experiments and theoretical background

In this Appendix we provide a review of literature that focuses on prior experiments that influenced our experimental strategy to study social trust. The core instrument of our experimental strategy is the investment game developed by Berg et al. (1995). Their experiments were designed “to study trust and reciprocity in an investment setting” (p122), starting a rich stream in the economic behavior literature. The typical experimental design of the IG involves a one-shot game under conditions of anonymity, where some subjects (first players or ‘trustors’) receive an endowment which they are invited to share with other subjects (second players or ‘trustees’), understanding that the shares they transfer will be tripled by the experimenter. Recipients may return any fraction (including zero or the full amount they receive) to the sender. Display of trust by the first player (sending money) and trustworthiness by the second player (returning money) can benefit both players. While trust and reciprocity can lead to mutual gain, the economic model based on the assumption that only self-interest motivates behavior predicts that those who receive the allocation will not send any positive amount, knowing that recipients of the money they send have no incentive to reciprocate and send back any positive amount, first players will not send any money (i.e., the subgame perfect equilibrium). Any positive sums sent by first players may reflect the consequences of social trust. This is so, since anonymity and having subjects play the game only once eliminate the possibility of strategically induced reciprocity motivated by fear of retaliation or fear of loss of reputation for trustworthiness in future games.

Since the publication of Berg et al. (1995), results of more than 200 replications of the IG in at least 35 countries have been reported in the literature (Johnson and Mislin, 2011). Most of these studies suggest that a significant number of subjects sent a positive sum of money to their anonymous partners in the IG and many recipients returned some of the funds they received to the senders. Many researchers considered the sums sent as measures of trust based on beliefs in the trustworthiness of the second player and the sums returned as measures of trustworthiness. The implicit assumption in most studies was that every participant in their ‘game’, irrespective of their diverse cultural, demographic, economic, and social backgrounds, interprets and plays this game as one where decisions are motivated by beliefs of the first player in the

trustworthiness of the second player and the response of the second player as manifestation of the player’s trustworthiness.

Given the strong evidence of context-specific influences and diverse preferences of people in economic games (Andreoni et al., 2003; Köszegi and Rabin, 2006; Tversky and Simonson, 1993), this assumption is not realistic. Indeed Berg et al. (1995) reviewing the results of their IG recognized that different subjects may interpret the game they play differently. They found that the correlation between the amount sent and the amount returned in their IG (in a ‘no-history setting’) was very small ($R^2 = 0.01$) and statistically not different from zero, and that 12 out of 28 subjects did not reciprocate. The authors observed that “... subjects who did not reciprocate may not have interpreted the trustors’ behavior as initiating trust”. They and other researchers suggested that the decision to send money can be attributed, in part, to others-regarding preferences that are not conditional on particular expectations with regard to the behavior of others. Eckel and Wilson (2004), for example, confirmed that belief in the trustworthiness of partners was not the main influence on the decisions made by their subjects. They asked the participants the following open-ended question (p.462): “What does this situation remind you of?” Only 26% of the first players and 9% of the second players indicated that expectations with regards to trustworthiness of the other player were the prevailing concern, while 17% of the first players and 12% of the second players indicated risk.

Cox (2004) used a cross-subjects triadic design to identify expectations of trustworthiness and reciprocity behaviors as opposed to transfers, which result from other-regarding unconditional preferences such as altruism. His experiment provided evidence of all three behaviors.

Ciriolo (2007) introduced the distinction between conditional and unconditional trustworthiness to account for the impacts of inequity aversion.

Ashraf et al. (2006:194) extended the effort to decompose money sent by trustors and returned by trustees to their underlying behavioral components. Each subject in their experiments participated in four games: a regular dictator game, a modified dictator game with the tripling of the amount sent and two investment games (once in the role of the first player and once in the role of a second player). The subjects also had to indicate choices among six gambles and certain amounts, as well as fill out a questionnaire that asked for demographic characteristics and trust attitudes. Their results suggested that trust (i.e., the willingness of a person to make himself/herself vulnerable to another) is based on beliefs in the other’s trustworthiness and unconditional others-regarding preferences such as altruism. Trustworthiness was shown to be based on unconditional other-regarding preferences and reciprocity.

Similarly, Johansson-Stenman et al. (2013), in field experiments conducted in Bangladesh, reported that slightly less than one-third of those who sent money expected to gain financially from their actions. They argued that “the motives behind the revealed behavior in trust games are more complicated and mixed than often believed” (p.287). They observed that some of the motives they uncovered may be outside the experimental context and “[t]his is something that could hardly have been observed from revealed behavior alone” (p.297). Indeed, while the question of what are the relationships of subjects’ stated motives, beliefs, and behavior in economic experiments have been the subject of a growing literature (e.g., trust - Johansson-Stenman et al., 2013; risk - Eckel and Wilson, 2004; Fetschenhauer and Dunning, 2012), the relationships between stated motives in the IG and behavior, to our knowledge, have not been tested until our study. Another innovation we have introduced was relaxation of the assumption of complete anonymity by introducing as manipulation condition of partial information about identity of second player (gender or age group).

Appendix II Overview of rick and dictator games results

In the risk game, the minimum, maximum, and average investments were \$4, \$20, and \$13.87, respectively. About 40% of players’ investments were less than or equal to \$10. There was a wide variation in

average investment among five social groups – mature people invested most (\$15.48) while elders invested the least (\$10.82), the average investment by males was \$15.07, females - \$12.59, and young - \$13.63. In all six cases of information about identity, average investment in RG was significantly higher (at a 5% significance level) than the money allocated by the first mover to the second mover in IG.

In the dictator game, in the case of no information about the second player, the average allocation was \$6.62 (33% of the pie), and average allocations by different social groups were: \$4.33 (21.6%) by young, \$8.2 (41%) by mature players and elders, \$7.41 (37%) by females and \$5.83 (29.2%) by males. In the cases of the second player being a male, a female, young, mature and elder, the average allocations were \$7.56, \$9.10, \$7.29, \$8.07, and \$11.60, respectively. The average allocations to males, females, mature players, and elders were significantly (at the 5% significance level) higher than the average allocation in the case of no information about the second player.

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